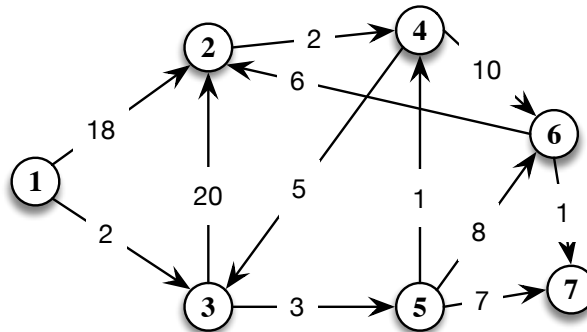


First part : Closed book

Total score: 21 pts. Minimum score to pass: 11 pts. Tentative scores: 1) 4 pts, 2) 5 pts, 3) 5 pts, 4) 3 pts.

1. Consider the directed graph in the picture. Compute the shortest path tree with root in node 1. Specify what is the best algorithm in the specific case of the type of graphs as that in the picture. Illustrate the intermediate steps. Let us suppose that the length of arc (4,6) changes. What are the values of the length of (4,6) that do not change the solution.



2. Consider the following linear programming problem:

$$\begin{aligned} \max \quad & -3x_1 \\ & -x_1 + x_2 \leq 4 \\ & -2x_1 - x_2 \leq 2 \\ & -2x_1 + x_2 \leq 4 \end{aligned}$$

Consider the two subsets of inequality indices: $B_1 = \{1, 2\}$ and $B_2 = \{2, 3\}$. For each of them discuss the primal and the dual feasibility conditions.

3. Consider the following binary knapsack problem:

$$\begin{aligned} \max \quad & 12x_1 + 7x_2 + 3x_3 + 3x_4 + 2x_5 - 2x_6 + 2x_7 \\ & 9x_1 + 5x_2 + 2x_3 + 3x_4 + 3x_5 + x_6 - 3x_7 \leq 8 \\ & x_i \in \{0, 1\}, i = 1, \dots, 7 \end{aligned}$$

Solve the problem by applying a branch and bound that uses the linear relaxation as upper bound. Compute an initial feasible solution by applying the greedy algorithm. Report the intermediate steps and the enumeration tree. When possible, use dominance rule to fix variables.

4. Consider the pair of dual problems $\min cx : Ax \leq b, x \in R^n$ and $\max yb, yA = c, y \leq 0, y \in R^m$. Prove that, given two feasible solutions $\bar{x}$ and $\bar{y}$, we have that $c\bar{x} \geq \bar{y}b$.

AMPL questions: 4 points.

in the rear of this paper

Write the definition in the ampl modelling language of the following:

- a set I .
- a set J .
- a parameter b_i with $i \in I$.
- a parameter c_j with $j \in J$.
- a non-negative integer variable $x_{i,j}$ with $i \in I$ and $j \in J$.

Then write an objective function that minimizes the following quantity:

$$\sum_{i \in I} \sum_{j \in J, c_j \neq b_i} (c_j b_i x_{i,j})$$

0.1 solution

```
set I;  
set J;  
param b{I};  
param c{J};  
var x{I,J} integer, >= 0;
```

```
minimize obj:  
  sum{i in I} sum{j in J: b[i] <> c[j]} (b[i]*c[j]*x[i,j]);
```

ANSWER IN THE SPACE BELOW